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SUPERSEDING
MIL-C-915/70(SHIPS)
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MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS, FOUR CONDUCTOR, TYPE FNW

(A) This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required.

Construction

First - Copper conductor, uncoated. (See details for size.)
Second - Separator, polyester type G of MIL-I-631, opaque white, applied either helically or longitudinally to give 100 percent coverage.
(A) Third - Cross-linked thermosetting polyethylene insulation, colored black. See details for thickness. Standard Identification Code applied by Method 1.
Fourth - Four conductors cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm, well-rounded assembly.
Fifth - Binder tape of 0.002 inch thick, corrugated polyester, type G of MIL-I-631.
Sixth - Standard thermoplastic jacket (gray). (See details for thickness.) Cable surface marking required.

Details

Type and size	Conductor Navy Standard size	Insulation thick. Nominal (inch)	Cable jacket thick. nominal (inch)	Overall diameter maximum (inches)	Conductor resistance per 1000 ft. at 25°C maximum (ohms)	Insulation resistance/ 1000 ft. minimum (megohms)	Voltage withstand minimum (volts)	Cold bending mandrel diameter maximum (inches)	Accelerated service loading current (amperes)
FNW-3	3(7)	0.030	0.040	0.447	4.00	6000	4000	4	---
FNW-4	4(7)	.030	.040	.513	2.51	5000	4000	4	---
FNW-9	9(7)	.030	.040	.630	1.25	3800	4000	5	---
FNW-23	23(7)	.045	.050	.830	.496	3700	5500	8	115

EXAMINATION AND TESTS:

Requirements:

Basic electrical:
Conductor resistance - Ohms/1000 ft., maximum . . . . . See details
Voltage withstand - Volts, rms, minimum . . . . . See details
Insulation resistance - Megohms/1000 ft., minimum . . . . . See Details

Group A:
Visual and dimensional

(A) Denotes changes

FSC 6145

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	Requirements:
Group B:	
Cold bending, cable - (See details for mandrel diameter)	No failure
Physicals (unaged)	
Insulation	
Tensile strength - Psi, minimum	1800
Elongation - Percent, minimum	250
Jacket	
Tensile strength - Psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Permanence of printing (conductor) - Cycles, minimum	25
Permanence of printing (jacket) - Cycles, minimum	250
Cable filler removability	2
Flammability - Inches, maximum	30
(A) Heat distortion - Percent of unaged, maximum	Zero
(A) Drip - At 95° ± 1°C.	No failure
(A) Accelerated service - (type FNW-23)	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - Percent of unaged, minimum	75
Elongation - Percent of unaged, minimum	75
Jacket	
Hot oil immersion	
Tensile strength - Psi, minimum	1400
Elongation - inches, minimum	1

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C plus the following:

(A) Cold working (minus 20° + 2°C.)	No damage
(A) Cable aging and compatibility (125° ± 5°C.)	No failure

UNIT ORDERING LENGTHS:

Type and size:	Feet (nominal)
FNW-3	2500
FNW-4, 9	2000
FNW-23	1500

(A) Custodians:
Army - MI
Navy - SH
Air Force - 80

(A) Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

(A) User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0609-50)

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